

# Technical Learning Report: Day 5

- **Date:** September 7, 2026
- **Intern Name:** Shekhar Singh Rawal
- **Subject:** Week 1 Revision, Test Planning & Estimation

## 1. Objective

The objective of Day 5 was to revise the major concepts covered during Days 1–5, with greater emphasis on **SRS, Test Plan, and Test Strategy**. The day also focused on understanding basic **test estimation techniques**, including Work Breakdown Structure, Three-Point Estimation, and Function Point Method. In addition, Selenium was introduced as a future automation testing topic, and Java was installed as preparation for upcoming Selenium learning.

## 2. Technical Breakdown

### A. Week 1 Revision

A short revision was conducted covering the major topics learned during the first five days, including:

- Software Testing fundamentals.
- SDLC models and test levels.
- Requirement analysis.
- Business, Functional, and UI rules.
- Static review techniques.
- SRS and requirement understanding.
- Test planning and test strategy.

### B. SRS, Test Plan and Test Strategy

Greater emphasis was given to understanding the relationship between **SRS, Test Plan, and Test Strategy**.

- **SRS:** Used as the source for understanding system requirements and expected system behavior.
- **Test Plan:** Defines what will be tested, the testing scope, approach, resources, schedule, risks, and testing activities.

- **Test Strategy:** Defines the overall approach and direction for testing, including how testing will be performed.

The revision helped connect requirements with test planning and the overall testing approach.

### C. Test Estimation Techniques

Three estimation approaches were studied:

1. **Work Breakdown Structure (WBS)**

Breaking a testing activity into smaller tasks so that individual activities can be identified and estimated.

2. **Three-Point Estimation**

Estimation using three values:

- Optimistic estimate
- Most likely estimate
- Pessimistic estimate

3. **Function Point Method**

An estimation approach based on the functional size of the application.

### D. Selenium and Java

Selenium was introduced as a topic for future automation testing learning.

Java was installed as preparation for learning and working with Selenium in the upcoming days.

## 3. Practical Implementation

### A. DemoQA Book Store Application

The **DemoQA Book Store Application** was selected for practical testing and learning.

- **Application:** DemoQA Book Store Application
- **Module:** Login

The application and Login module were used as the practical context for connecting requirements, test planning, and testing concepts.

### B. Test Documentation Revision

The Test Plan and Test Strategy were revised with greater focus on how they support testing of an application and how they relate to the requirements documented in the SRS.

## C. Estimation Practice

The three estimation techniques were studied and applied conceptually to understand how testing work can be divided and estimated using:

- Work Breakdown Structure.
- Three-Point Estimation.
- Function Point Method.

## D. Selenium Preparation

Java was successfully installed as preparation for future Selenium automation learning.

# 4. Key Findings & Best Practices

- SRS provides the requirements that form the basis for understanding what the application should do.
- The Test Plan provides a structured plan for testing the application.
- Test Strategy defines the overall direction and approach to testing.
- Test estimation helps break testing work into manageable activities and determine the effort required.
- WBS focuses on breaking work into smaller tasks.
- Three-Point Estimation considers optimistic, most likely, and pessimistic estimates.
- Function Point Method focuses on the functional size of the application.
- Selenium was identified as a future automation testing topic.
- Java was installed as preparation for upcoming Selenium learning.

## 5. Conclusion

Day 5 provided a revision of the major concepts covered during Days 1–5, with particular emphasis on **SRS, Test Plan, and Test Strategy**. The day also introduced three test estimation techniques: **Work Breakdown Structure, Three-Point Estimation, and Function Point Method**. Practical focus was given to the **DemoQA Book Store Application Login module**, while Java was installed to prepare for future Selenium automation testing activities.

## Glossary

| Term                       | Definition                                                                                                                        | Example                                                                                                    |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Acceptance Criteria</b> | Conditions that must be satisfied before the customer/user accepts the software.                                                  | Login is accepted only if a valid user can log in, invalid users get an error, and the password is hidden. |
| <b>Acceptance Testing</b>  | Testing done to check whether the software meets the user's needs and acceptance criteria.                                        | Customer tests the login page and confirms it works as requested before approving the system.              |
| <b>Actor</b>               | A person or another system that interacts with the software.                                                                      | In an online store, <b>Customer</b> , <b>Admin</b> , and <b>Payment Gateway</b> can be actors.             |
| <b>Actual Result</b>       | What actually happens when you run a test.                                                                                        | <b>Expected:</b> Error message appears for wrong password. <b>Actual:</b> No error message appears.        |
| <b>Alpha Testing</b>       | Testing performed by potential users or an independent test team at/near the developer's site before release to the wider public. | A software company gives an unfinished app to a small group of testers at its office for testing.          |
| <b>Audit</b>               | An independent check to see whether software processes/products follow required standards, rules, or procedures.                  | An auditor checks whether the company follows its required software testing process.                       |

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| <b>Audit Trail</b>                     | A record/path that allows you to trace what happened to data from its original input to its final output. | A bank transaction record shows <b>who made the transaction, when, what changed, and what the final result was.</b>                      |
| <b>Availability</b>                    | How much of the time a system is working and accessible when users need it.                               | If a website is available 99.9% of the time, its availability is <b>99.9%</b> .                                                          |
| <b>Best Practice</b>                   | A commonly accepted and effective way of doing something.                                                 | Using <b>code reviews and automated tests</b> before releasing software can be a best practice.                                          |
| <b>Beta Testing</b>                    | Real users/customers test the software in a real or external environment before the final release.        | A company releases a new app to 1,000 selected users so they can use it and report problems.                                             |
| <b>Black-Box Test Design Technique</b> | Designing test cases by looking at requirements/specifications without looking at the internal code.      | Requirement says password must be <b>8–20 characters</b> . Tester creates tests for 7, 8, 20, and 21 characters without seeing the code. |
| <b>Black-Box Testing</b>               | Testing the software from the outside without knowing or considering its internal code/implementation.    | Enter username/password → click Login → check whether login works. The tester doesn't need to know how the login code works.             |
| <b>Boundary Value</b>                  | A value at or very close to the edge of an allowed range.                                                 | If age must be <b>18–60</b> , boundary values include <b>17, 18, 19, 59, 60, 61</b> .                                                    |

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| <b>Boundary Value Analysis (BVA)</b> | A black-box technique where we test values at and around the boundaries because defects commonly occur there. | For a field allowing <b>1–100</b> , test <b>0, 1, 2, 99, 100, 101</b> . |
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